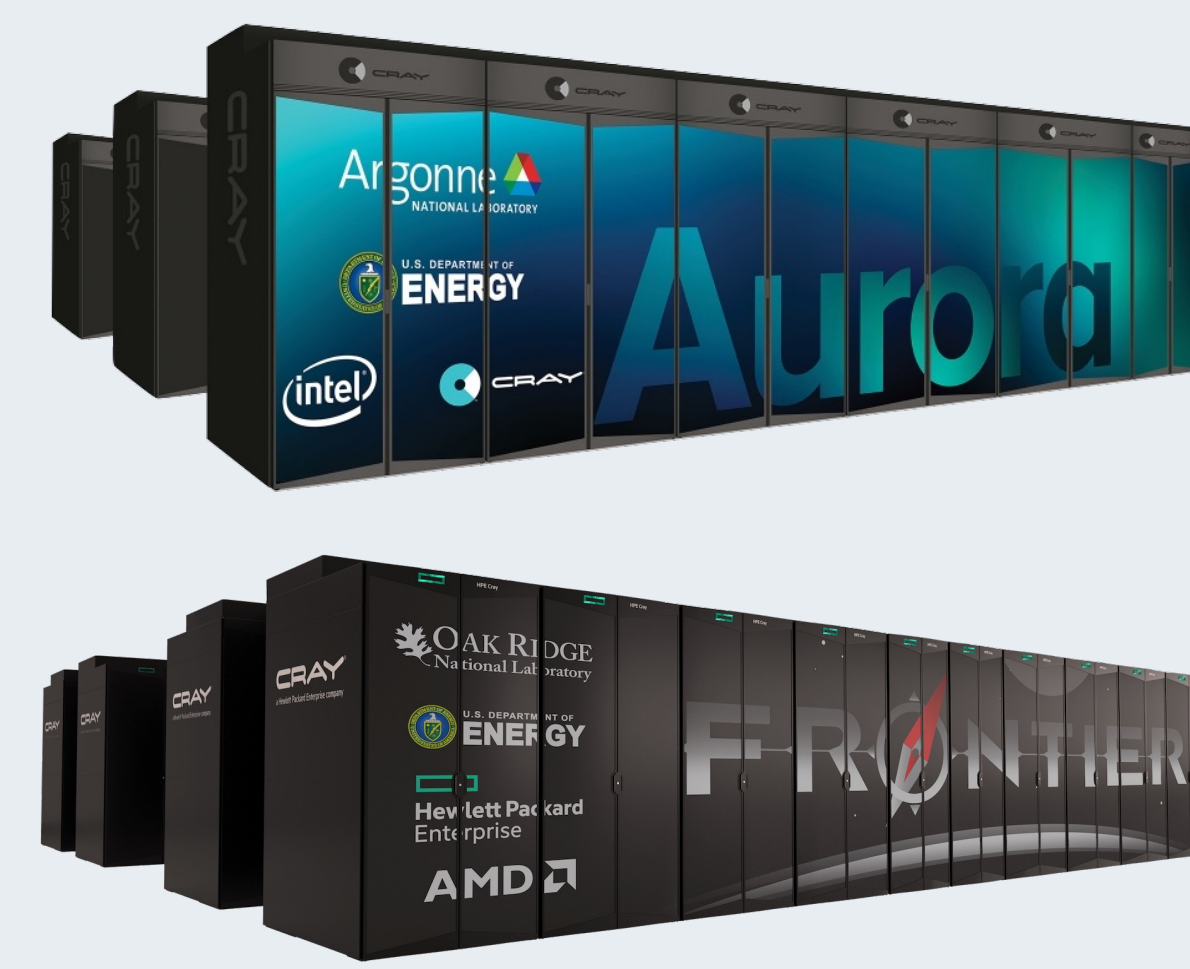


Solver-Integrated Lossy and Lossless Compression for Scalable Flow Simulations

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THE STORAGE AND I/O WALL IN EXASCALE COMPUTING

Modern GPU-accelerated HPC simulations routinely generate datasets ranging from hundreds of gigabytes to **multiple terabytes** per snapshot, making storage, I/O, and data movement major bottlenecks in exascale scientific workflows. As compute capability continues to scale, filesystem bandwidth and storage infrastructure increasingly limit output cadence, visualization, and downstream analysis.

BOTTLENECKS

- Multi-terabyte simulation snapshots
- Filesystem and MPI-I/O bottlenecks
- High storage and data transfer costs
- Limited output cadence at extreme scale
- Expensive post-hoc analysis workflows



Result: I/O and storage wall limits science and productivity

OUR SOLUTION: SOLVER-INTEGRATED COMPRESSION

We integrate both **lossless** compression for restart/checkpoint reliability and **error-bounded lossy compression** for analysis-oriented outputs directly into the GPU-enabled spectral-element solver nekRS. By compressing solution fields in situ before MPI-I/O, we substantially reduce I/O volume and storage cost while preserving scientific value.

BENEFITS

- Reduces storage footprint and I/O volume
- Minimizes data movement overhead
- Preserves QoIs within controlled tolerances
- Enables higher output cadence at extreme scale
- Maintains scientific usability of reconstructed fields



Result: Lower storage & I/O cost, higher output cadence, and scalable science

METHODOLOGY: COMPRESSION-AWARE I/O FRAMEWORK

A portable, compression-aware I/O framework integrated in CFD solver



HPC Solver (nekRS)

- Spectral-element discretization
- High-order accuracy
- GPU-accelerated



Rank-local Fields (Element-based)

- Each rank holds a subset of elements
- No communication for compression



Compression (in situ)

- Error-bounded SZ3 for analysis
- Lossless Blosc2 for checkpoints



Portable Compressed Format

- Rank-decomposable metadata
- Decouples storage from MPI layout
- Supports arbitrary read-time parallelism



Readers & Workflows

- Visualization · Post-hoc analysis
- ROM · ML / GNN pipelines
- In situ or ex situ access



Rank-local data reduction preserving portability across downstream workflows

QoI PRESERVATION

Guaranteed pointwise error bound $|u_i - \tilde{u}_i| \leq \varepsilon_{rel} \cdot \Delta u_{range}$ or ε_{abs}
Closed form a priori bounds on physical QoI

Define: $\eta = \varepsilon_{rel} \cdot (\Delta u_{range})^{max}$ or ε_{abs} , Pick ε from target accuracy

QoI	Formula	Error-bound scaling, small η
Mean profiles	$\langle u \rangle$	η
RMS / variance	$\langle u'^2 \rangle = \langle u^2 \rangle - \langle u \rangle^2$	$4\eta \cdot \max(u)$
Velocity magnitude	$ u = \sqrt{u^2 + v^2 + w^2}$	$\sqrt{3} \eta$
Kinetic energy	$E_k = \frac{1}{2} u ^2$	$\sqrt{3} \eta \cdot \max(u , v , w)$
Turbulent KE	$k = \frac{1}{2} u' ^2$	$6\eta \cdot \max(u' , v' , w')$
Gradients vorticity, stress etc.	$\nabla u, \omega = \nabla \times u, \tau = \nabla u + (\nabla u)^T$	$O(\eta/h)$ grid-dependent

WORKED EXAMPLE: FUSION BLANKET VELOCITY

Given: $u_x, u_y, u_z \in [-58, 58]$, $\varepsilon_{rel} = 10^{-3}$, $\Delta u_{range} = 116$

Per-component bound: $|u_i - \tilde{u}_i| \leq 0.116$

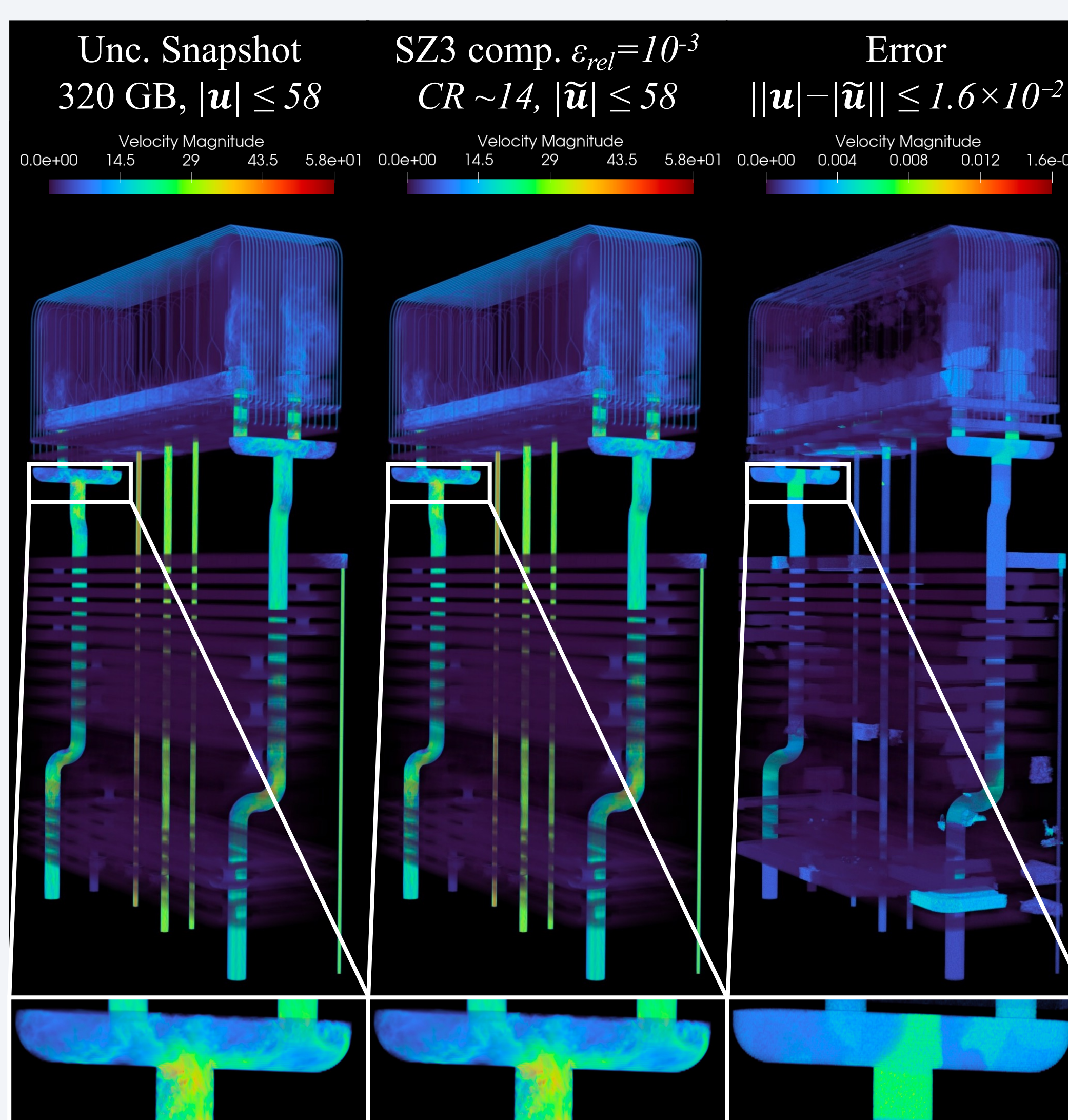
Magnitude bound: $\|u\| - \|\tilde{u}\|_{\infty} \leq \sqrt{3} \cdot \varepsilon_{rel} \cdot \Delta u_{range} \approx 0.20$ (worst-case)

Observed: $\|u\| - \|\tilde{u}\|_{\infty} \leq 1.6 \times 10^{-2} \sim 12 \times$ tighter than worst-case



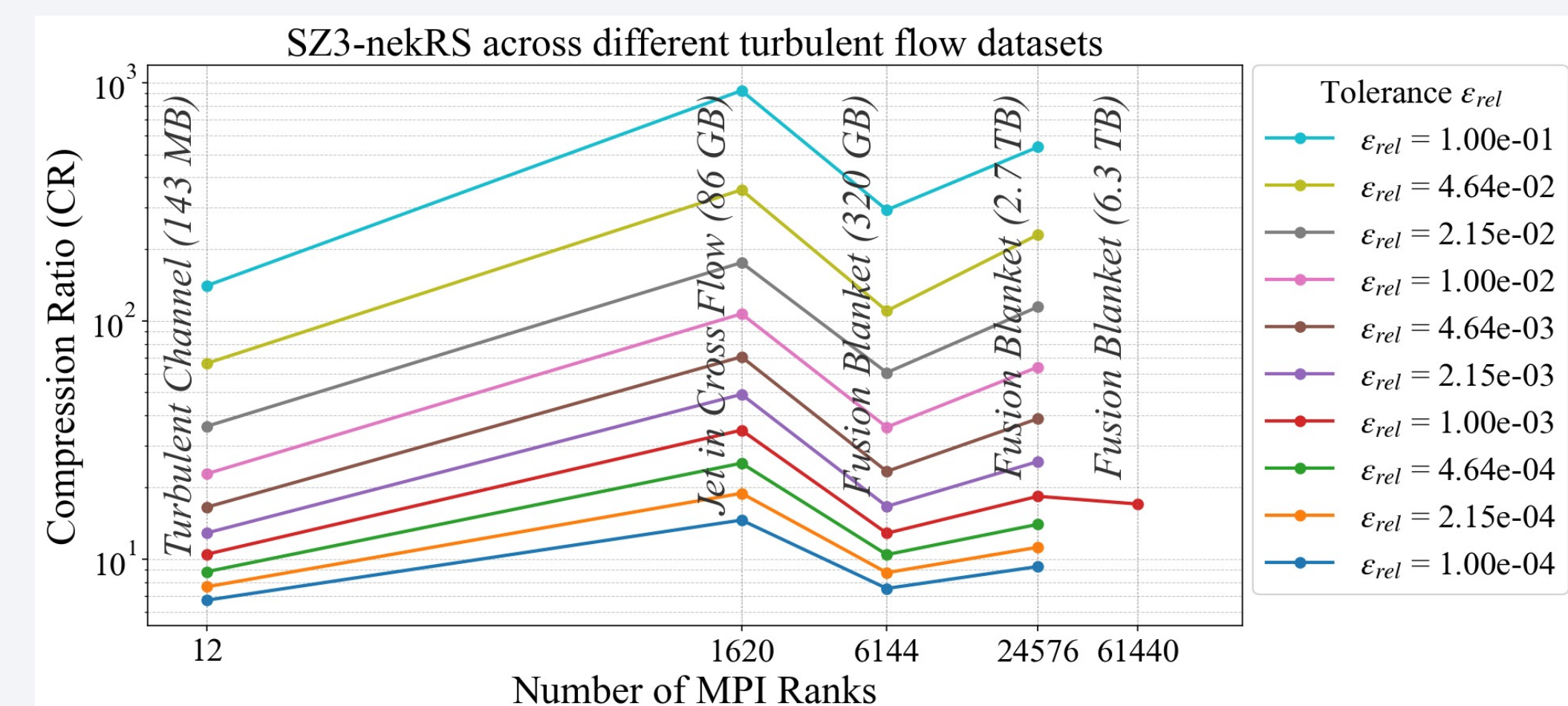
Pointwise tolerance translates to QoI guarantees

FUSION BLANKET: VELOCITY MAGNITUDE

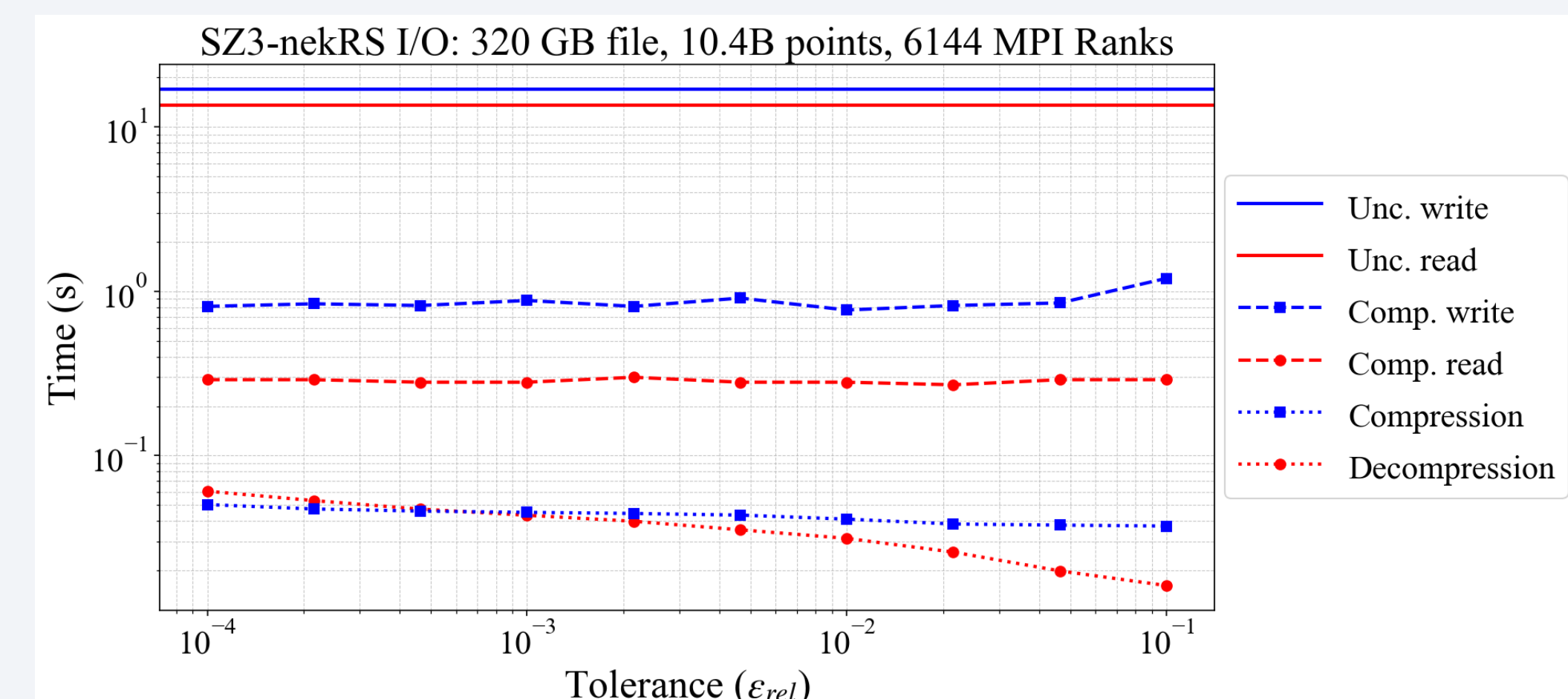


Key flow structures are preserved after error-bounded compression

PERFORMANCE: SCALING & I/O TIME



Scaling validated across multiple flow cases beyond 60,000 MPI ranks



10-100x compression ratio and up to 20-50x I/O time reduction

FUTURE WORK

- GPU-resident compression and asynchronous I/O
- SEM-aware transform-based compression strategies
- Integration with visualization tools (ParaView / VisIt)
- Evaluation of additional lossless and lossy compressors
- Additional directions:**
 - Reactive and multiphysics flows
 - Adaptive tolerance selection
 - Compressed ML-ready datasets for ROM and AI workflows

GENERALIZATION

- Applicable to a wide range of distributed scientific workflows with locally structured data layouts.
- High-order FEM / DG
- Structured-grid CFD
- Multiphysics workflows
- Key ideas:**
 - Rank-local, in situ data reduction
 - QoI-aware error-bounded compression
 - Compressed parallel MPI-I/O



A general strategy for scalable, data-centric HPC workflows

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- Generative AI tools assisted with language editing and code/debugging tasks.



Data reduction is as critical as numerical computation in the exascale era

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